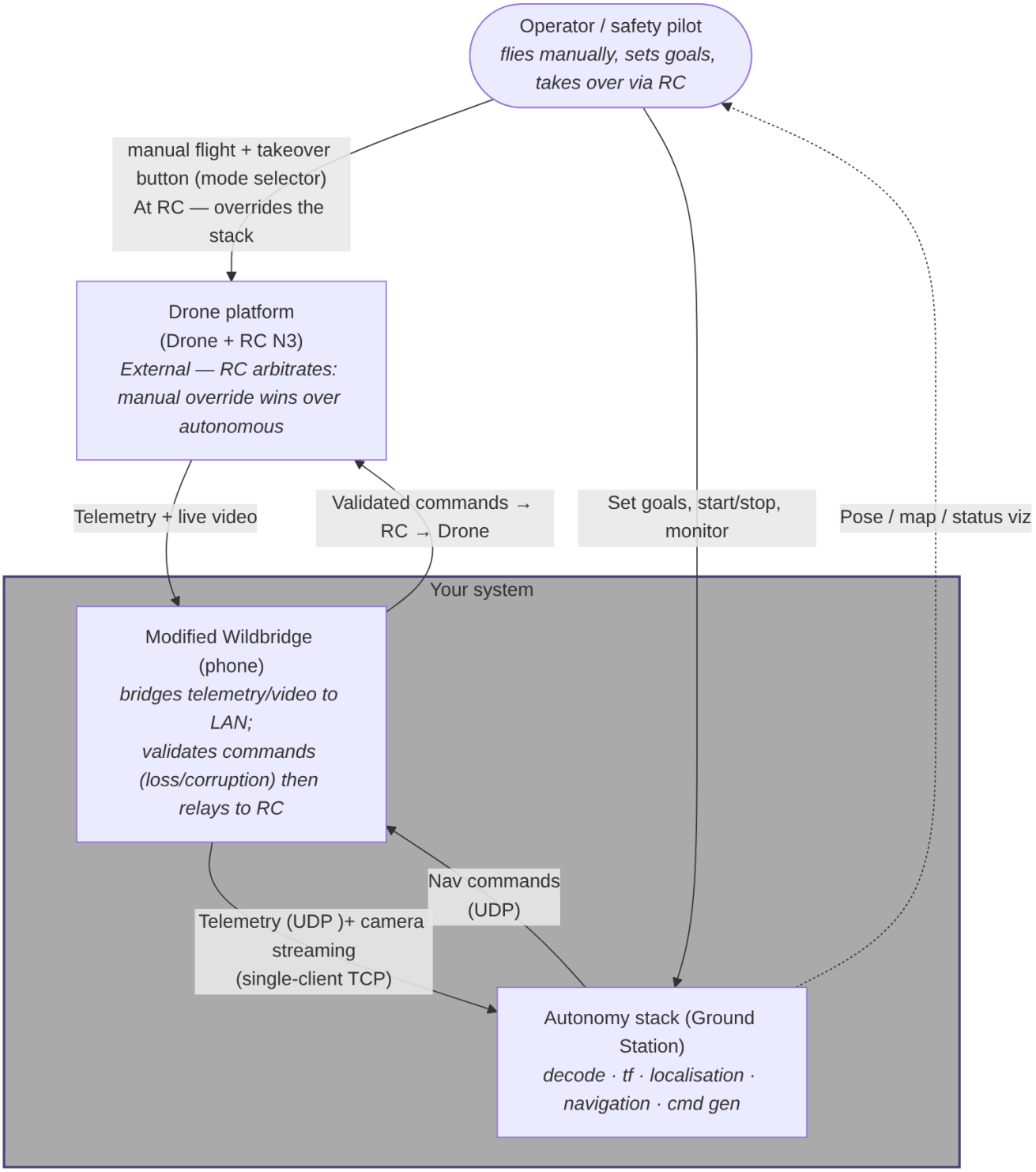
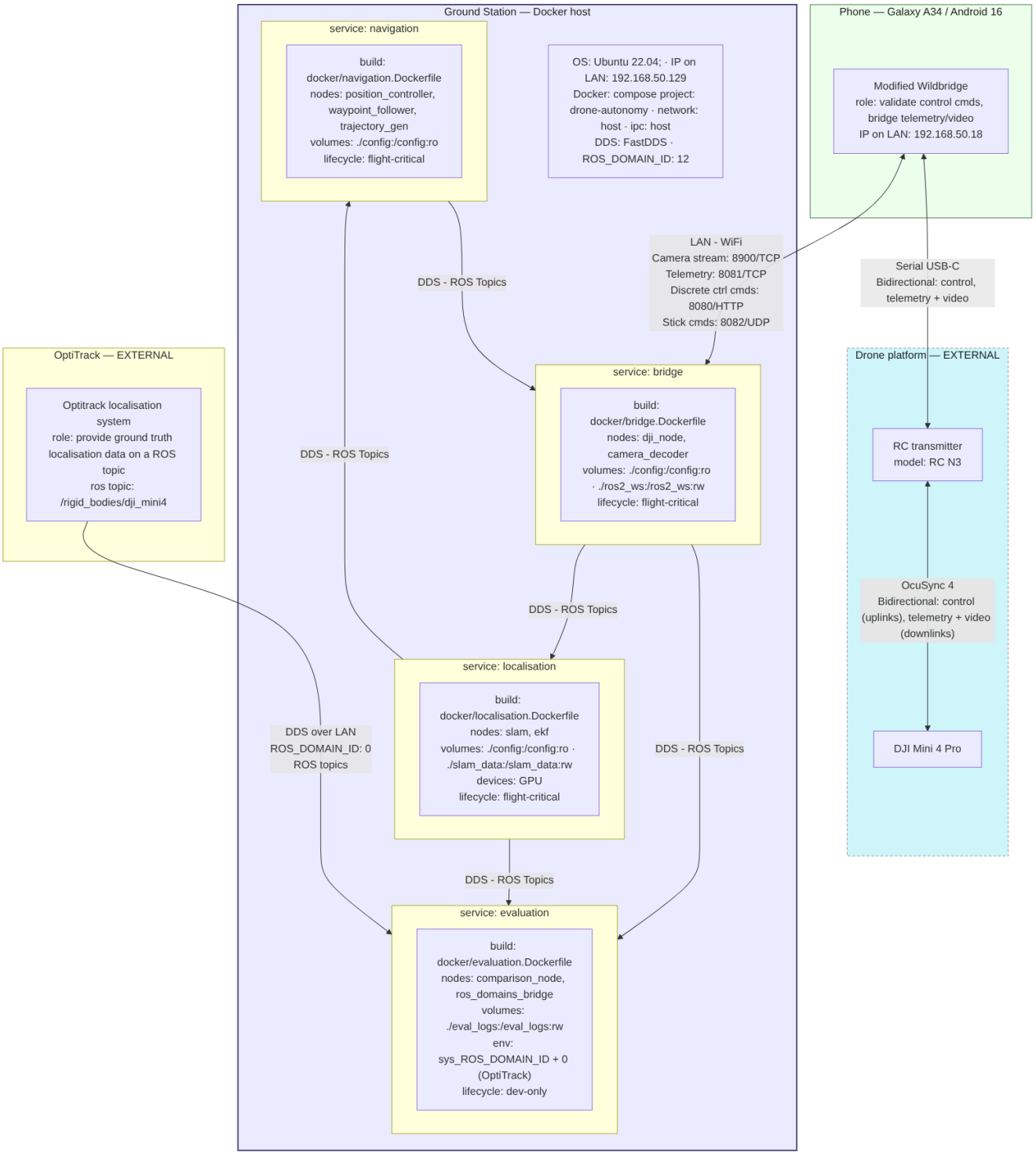
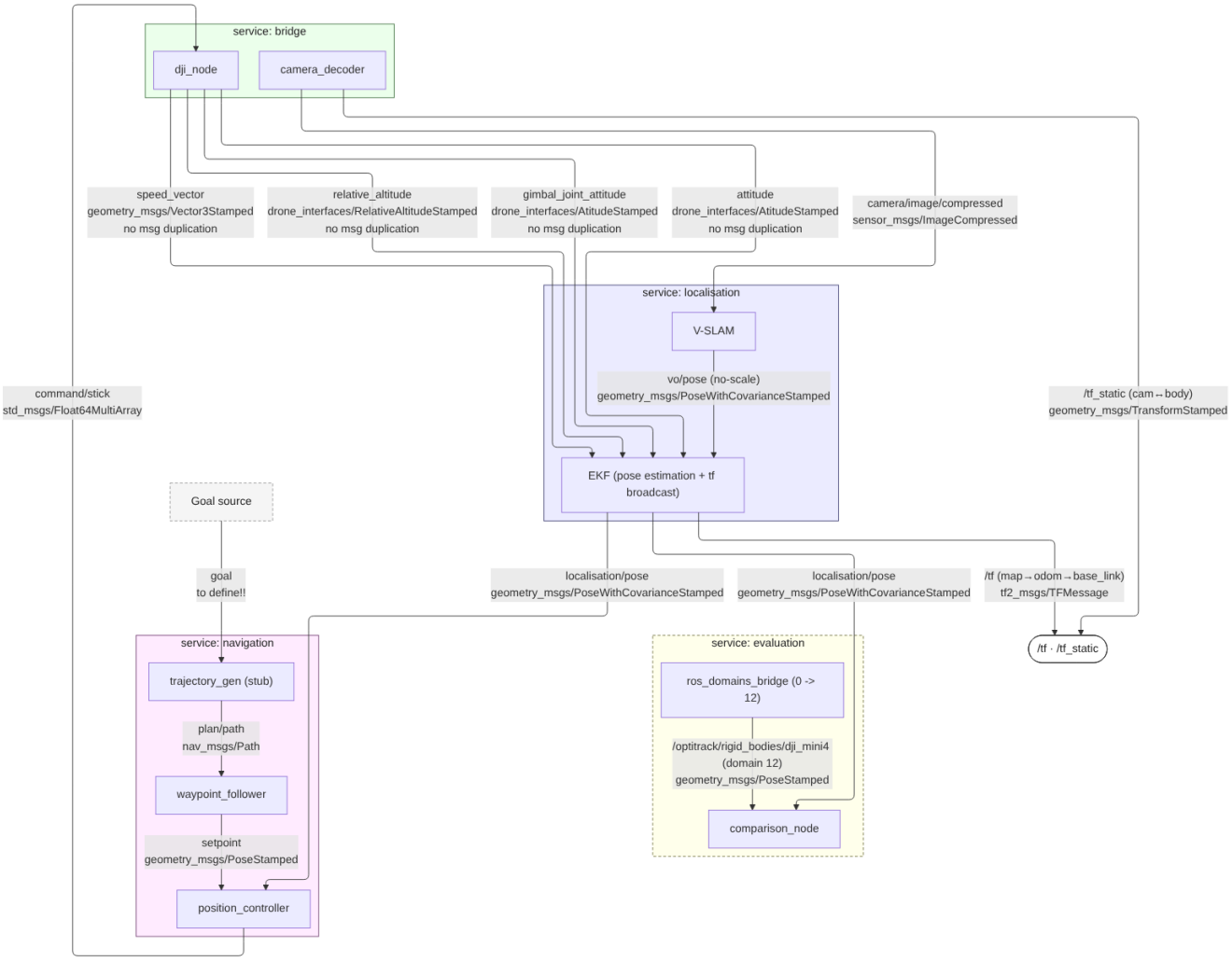
system context



deployment


Ground Station — Docker host
service: navigation
build:
docker/navigation.Dockerfile
nodes: position_controller,
waypoint_follower,
trajectory_gen
volumes: ./config:/config:ro
lifecycle: flight-critical
OS: Ubuntu 22.04; · IP on
LAN: 192.168.50.129
Docker: compose project:
drone-autonomy · network:
host · ipc: host
DDS: FastDDS ·
ROS_DOMAIN_ID: 12
Phone — Galaxy A34 / Android 16
Modified Wildbridge
role: validate control cmds,
bridge telemetry/video
IP on LAN: 192.168.50.18
LAN - WiFi
Camera stream: 8900/TCP
Telemetry: 8081/TCP
Discrete ctrl cmds:
8080/HTTP
Stick cmds: 8082/UDP
Serial USB-C
Bidirectional: control,
telemetry + video
DDS - ROS Topics
OptiTrack — EXTERNAL
Optitrack localisation
system
role: provide ground truth
localisation data on a ROS
topic
ros topic:
/rigid_bodies/dji_mini4
service: bridge
build:
docker/bridge.Dockerfile
nodes: dji_node,
camera_decoder
volumes: ./config:/config:ro
· ./ros2_ws:/ros2_ws:rw
lifecycle: flight-critical
Drone platform — EXTERNAL
RC transmitter
model: RC N3
DDS - ROS Topics
OcuSync 4
Bidirectional: control
(uplinks), telemetry + video
(downlinks)
DDS - ROS Topics
service: localisation
build:
docker/localisation.Dockerfile
nodes: slam, ekf
volumes: ./config:/config:ro ·
./slam_data:/slam_data:rw
devices: GPU
lifecycle: flight-critical
DJI Mini 4 Pro
DDS over LAN
ROS_DOMAIN_ID: 0
ROS topics
DDS - ROS Topics
DDS - ROS Topics
service: evaluation
build:
docker/evaluation.Dockerfile
nodes: comparison_node,
ros_domains_bridge
volumes:
./eval_logs:/eval_logs:rw
env:
sys_ROS_DOMAIN_ID + 0
(OptiTrack)
lifecycle: dev-only

node graph


service: bridge
dji_node
camera_decoder
speed_vector
geometry_msgs/Vector3Stamped
no msg duplication
relative_altitude
drone_interfaces/RelativeAltitudeStamped
no msg duplication
gimbal_joint_attitude
drone_interfaces/AtitudeStamped
no msg duplication
attitude
drone_interfaces/AtitudeStamped
no msg duplication
camera/image/compressed
sensor_msgs/ImageCompressed
service: localisation
V-SLAM
vo/pose (no-scale)
geometry_msgs/PoseWithCovarianceStamped
EKF (pose estimation + tf broadcast)
command/stick
std_msgs/Float64MultiArray
/tf_static (cam↔body)
geometry_msgs/TransformStamped
Goal source
goal
to define!!
localisation/pose
geometry_msgs/PoseWithCovarianceStamped
localisation/pose
geometry_msgs/PoseWithCovarianceStamped
/tf (map→odom→base_link)
tf2_msgs/TFMessage
/tf · /tf_static
service: navigation
trajectory_gen (stub)
plan/path
nav_msgs/Path
waypoint_follower
setpoint
geometry_msgs/PoseStamped
position_controller
service: evaluation
ros_domains_bridge (0 -> 12)
/optitrack/rigid_bodies/dji_mini4 (domain 12)
geometry_msgs/PoseStamped
comparison_node

tf tree

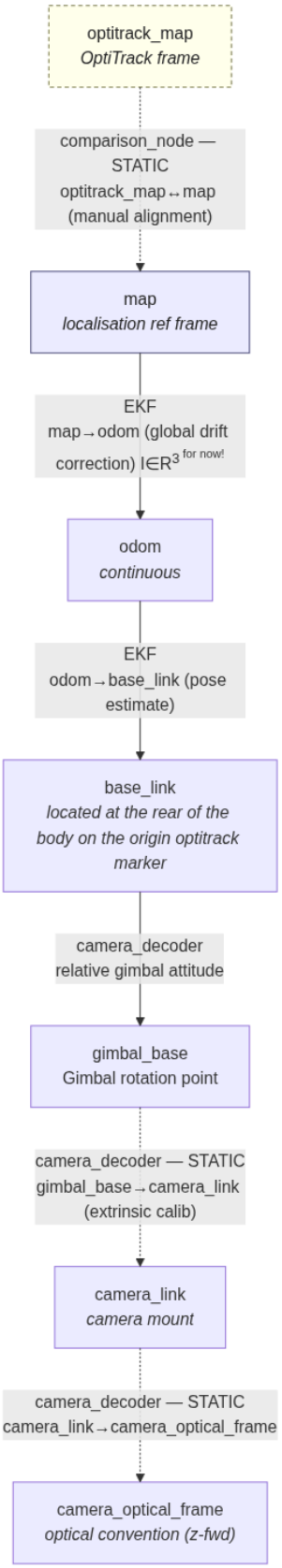

optitrack_map
OptiTrack frame
comparison_node — STATIC
optitrack_map ↔ map
(manual alignment)
map
localisation ref frame
EKF
map → odom (global drift correction) l∈R3 for now!
odom
continuous
EKF
odom → base_link (pose estimate)
base_link
located at the rear of the body on the origin optitrack marker
camera_decoder
relative gimbal attitude
gimbal_base
Gimbal rotation point
camera_decoder — STATIC
gimbal_base → camera_link
(extrinsic calib)
camera_link
camera mount
camera_decoder — STATIC
camera_link → camera_optical_frame
camera_optical_frame
optical convention (z-fwd)

Diagram 5 — Node Interface Specification

Black-box interface for each node.

Convention: parameters are DECLARED in-node with defaults; tuned values live in `config/<node>/params.yaml`.

Service: bridge

dji_controller dji_node (adopted from WildBridge)

Job: bidirectional bridge between the modified WildBridge and ROS over TCP/UDP.

Direction	Topic	Type	Notes
pub	`speed_vector`	`geometry_msgs/Vector3Stamped`	stamp with measurement time
pub	`relative_altitude`	`Relative AltitudeStamped`	Height above launch, from KeyAltitude
pub	`gimbal_joint_attitude`	`drone_interfaces/AttitudeStamped`	Gimbal attitude (header, {"pitch", "roll", "yaw"})
pub	`attitude`	`drone_interfaces/AttitudeStamped`	Drone's atttiude (header, {"pitch", "roll", "yaw"})
sub	`command/stick`	`std_msgs/Float64MultiArray`	[lx, ly, rx, ry]

Params: `phone_IP: 192.168.50.18`, `tcp_port: 8081`, `udp_port: 8082`, `http_port: 8080`, `telemetry publish rate: Measured 10 Hz FC / 19 Hz gimbal`.

camera_streamer camera_decoder

Job: receive the encoded camera stream (single-client TCP), decode, publish frames; broadcast the static camera extrinsic.

Direction	Topic	Type	Notes
pub	`camera/image_raw`	`sensor_msgs/Image`	decoded frame
pub	`/tf_static`	`geometry_msgs/TransformStamped`	base->gimbal->camera->optical
sub	`gimbal_joint_attitude`	`drone_interfaces/AtitudeStamped`	Gimbal joint states {"pitch", "roll", "yaw"}

Params: `camera_tcp_port: 8900`, `frame_ids: camera_link, camera_optical_frame`.

Service: localisation

localisation slam_node

Job: monocular visual odometry with scale ambiguity.

Direction	Topic	Type	Notes
sub	`camera/image_raw`	`sensor_msgs/Image`	
pub	`vo/odom`	`nav_msgs/Odometry`	no metric scale
pub	`localisation_status`	`std_msgs/String`	Tracking state

Params: `vocabulary: ORBvoc`, `camera calibration: config/camera_calibration.yaml`, `feature settings: <settings>`, `devices: CPU`.

localisation ekf_node

Job: fuse V-SLAM with metric velocity + relative altitude to resolves scale, outputs metric pose; broadcasts dynamic tf.

Direction	Topic	Type	Notes
sub	`vo/odom`	`nav_msgs/Odometry`	
sub	`speed_vector`	`geometry_msgs/Vector3Stamped`	metric velocity (scale ref)
sub	`relative_altitude`	`std_msgs/Float64`	vertical metric anchor
sub	`gimbal_joint_attitude`	`drone_interfaces/AttitudeStamped`	Gimbal attitude (header, {"pitch", "roll", "yaw"})
sub	`attitude`	`drone_interfaces/AttitudeStamped`	Drone's atttiude (header, {"pitch", "roll", "yaw"})
sub	`localisation_status`	`std_msgs/String`	Tracking state
pub	`localisation/pose`	`geometry_msgs/PoseWithCovarianceStamped`	source-agnostic pose
pub	`/tf`	`tf2_msgs/TFMessage`	map->odom->base_link

Params: `covariances`, `scale-init: <scale-init>`, `frame ids: odom, base_link`.

Service: navigation

navigation trajectory_gen_node (To be defined)

Job: turn a goal into a path/setpoint sequence. Stubbed until planner chosen.

Direction	Topic	Type	Notes
sub	`goal`	`TBD`	goal msg type to define
pub	`plan/path`	`nav_msgs/Path`	

Params: planner settings (TBD).

navigation waypoint_follower_node

Job: walk along the path, emit the current setpoint.

Direction	Topic	Type	Notes
sub	`plan/path`	`nav_msgs/Path`	
pub	`setpoint`	`geometry_msgs/PoseStamped`	current target

Params: `waypoint reach tolerance: <tolerance>`, `lookahead: <lookahead>`.

navigation position_controller_node

Job: closed-loop control. Given current pose + setpoint, produce stick velocity.

Direction	Topic	Type	Notes
sub	`localisation/pose`	`geometry_msgs/PoseWithCovarianceStamped`	current pose
sub	`setpoint`	`geometry_msgs/PoseStamped`	target
pub	`command/stick`	`std_msgs/Float64MultiArray`	[lx, ly, rx, ry]

Params: `PID gains: <gains>`, ...

Consumes tf for frame lookups.

Service: evaluation (dev-only, metrics)

evaluation ros_domains_bridge (domain_bridge)

Job: bridge the OptiTrack ground-truth topic from domain 0 into domain 12.

Direction	Topic	Type	Notes
sub	`/optitrack/rigid_bodies/dji_mini4` (domain 0)	`geometry_msgs/PoseStamped`	external, absolute
pub	`/optitrack/rigid_bodies/dji_mini4` (domain 12)	`geometry_msgs/PoseStamped`	into custom domain

Params: `ros domain ids: 0, 12` 0 -> Optitrack, 12 -> Stack.

evaluation comparison_node

Job: compare localisation estimate vs ground truth, compute + log ATE/RPE.

Direction	Topic	Type	Notes
sub	`localisation/pose`	`geometry_msgs/PoseWithCovarianceStamped`	our estimate
sub	`/optitrack/rigid_bodies/dji_mini4` (domain 12)	`geometry_msgs/PoseStamped`	ground truth
pub	`eval/metrics`	`TBD`	ATE/RPE

Params: `log path: evaluation/eval_logs`.

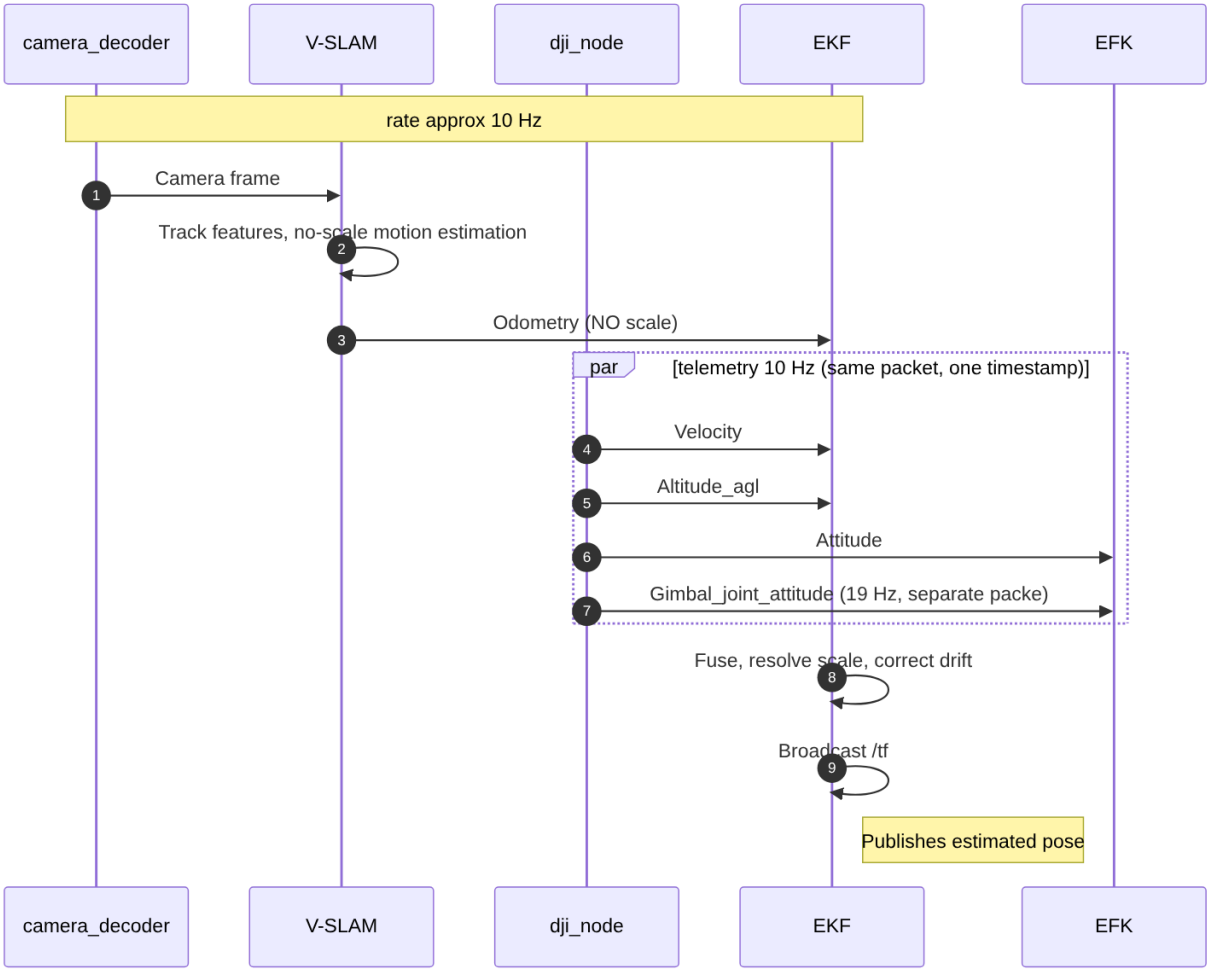
To be defined!!

- `goal` message type — undecided.
- `eval/metrics` — publish a msg, or log-to-file only?
- EKF: full `map->odom->base_link` chain vs collapsed `map->base_link`.
- `satellite_count` — published by dji_node but no consumer until the GNSS/visual switch exists.

Data Flow / Sequence

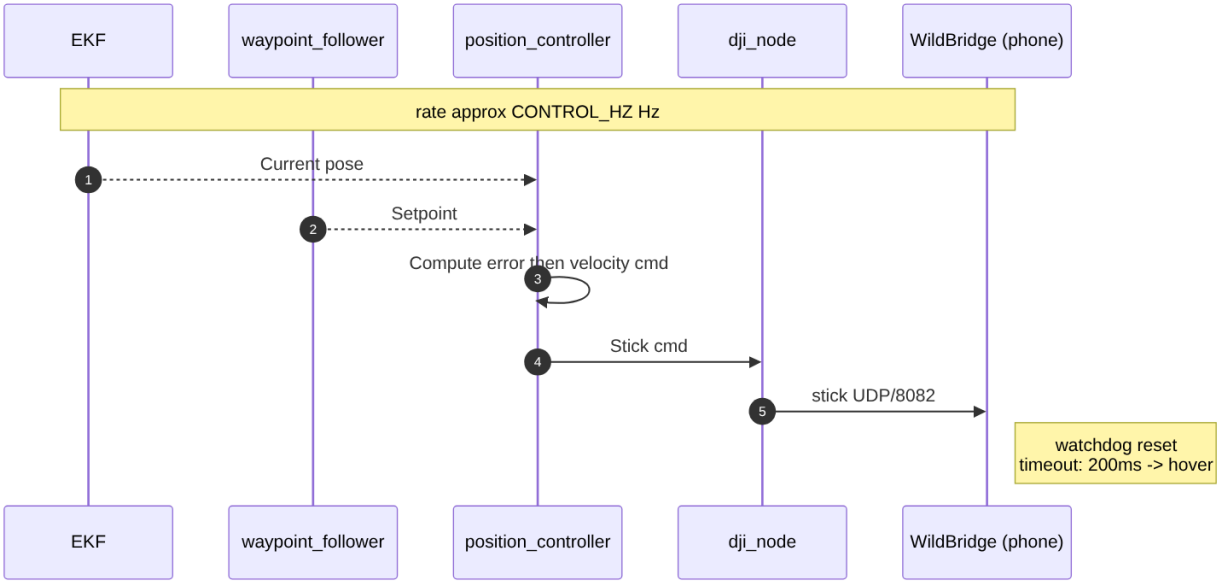
Traces frames, pose, velocity and commands through the nodes over time.

Localisation — runs at camera rate



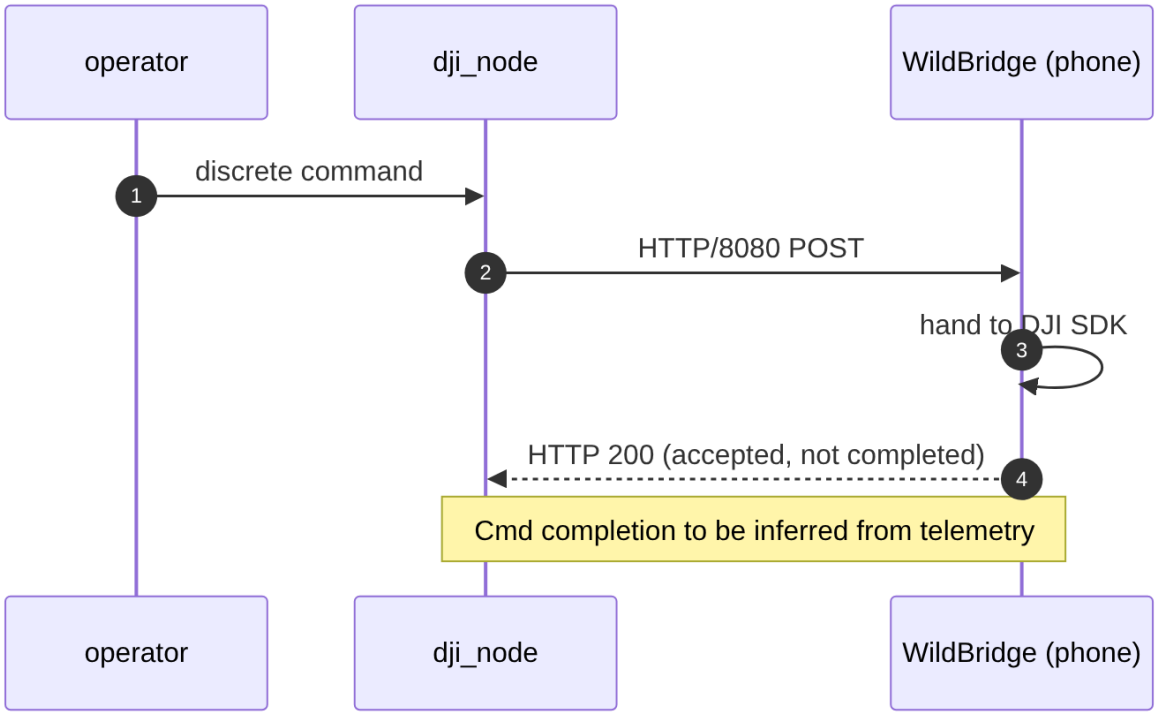
Timing note: telemetry carries measurement-time stamps (phone key listener + clock-offset mapping), so its latency is resolved. Camera frames are stamped at decode time minus VIDEO_LATENCY, still to be measured. Fuse in timestamp order.

6b. Control loop - ~high rate

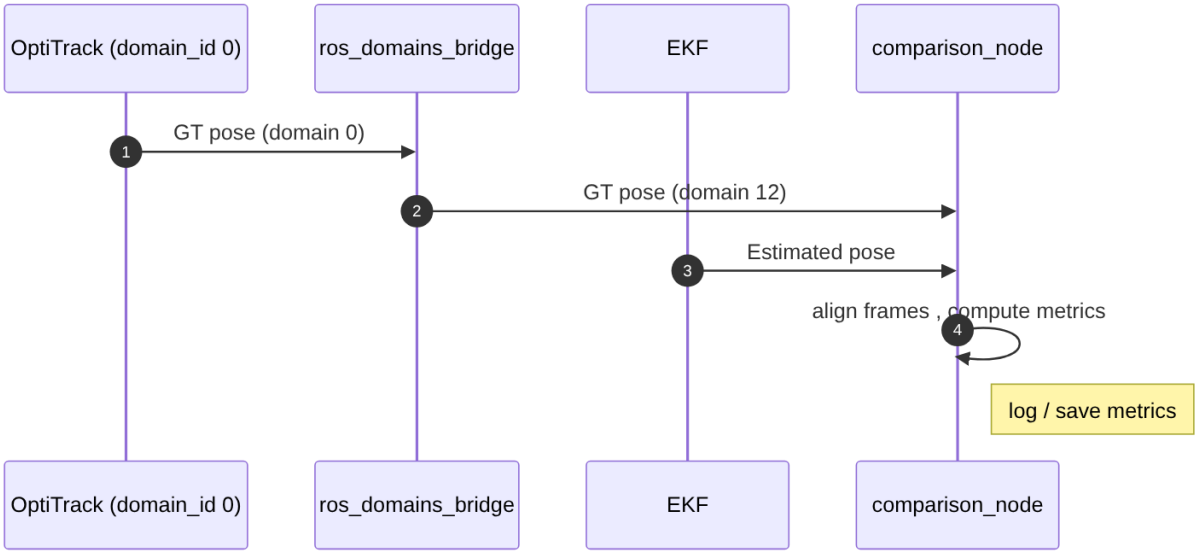


Timing note: pose age at the controller is ~310 ms (video ~200 + SLAM ~40 + gimbal tf ≤53 + filter ~10). This is not compared against the watchdog — the watchdog checks command freshness, not pose age. position_controller runs on a fixed timer and publishes regardless of pose arrival, so the watchdog is satisfied by construction. The 310 ms instead caps position-loop bandwidth at ~0.15 Hz, which is why waypoint_follower needs a conservative max_velocity.

6c. Discrete commands (takeoff / land) — event-driven, reliable



6d. Evaluation flow (dev-only) — low rate



Notes

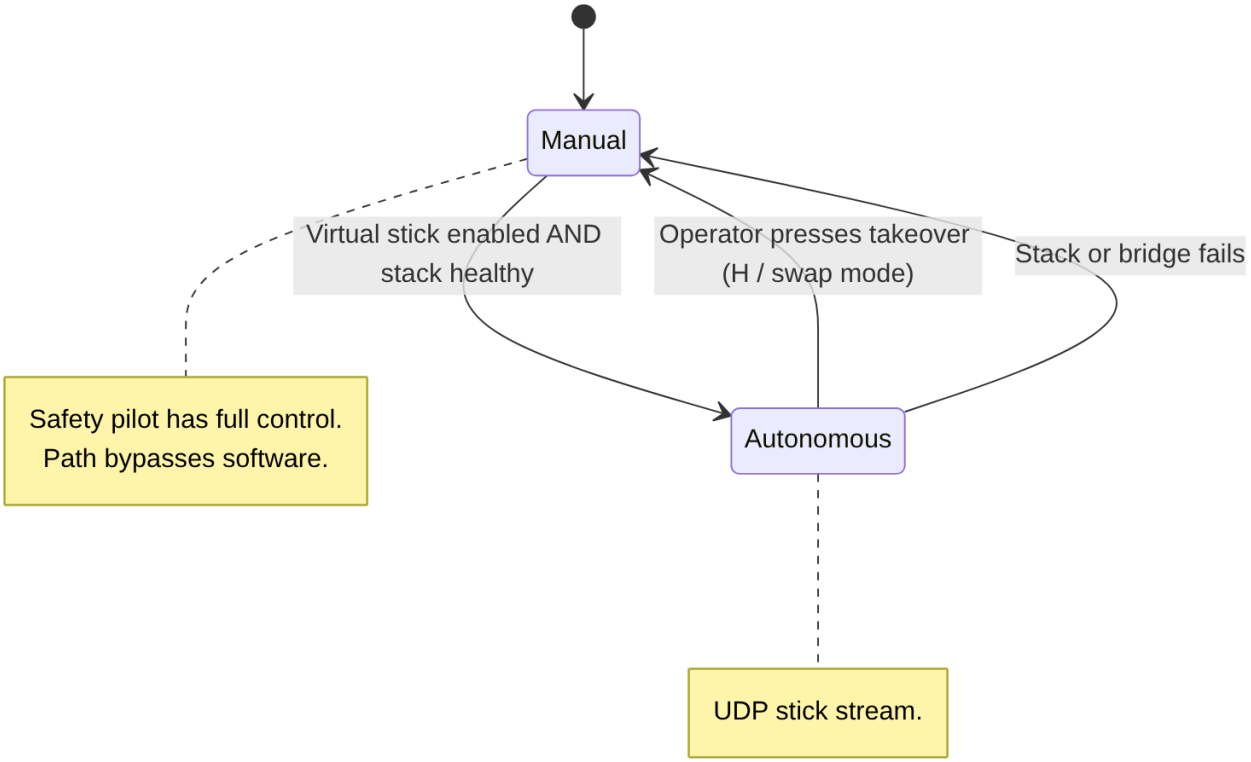
- 6b + 6c share the same drone command path, hoever 6b is continuous/loss-tolerant (UDP) and 6c is discrete/reliable (HTTP + ACK).

System State Machine (flight modes + safety)

Behavioural changes and the transitions between them. Also, it inlcudes the system safety logic.

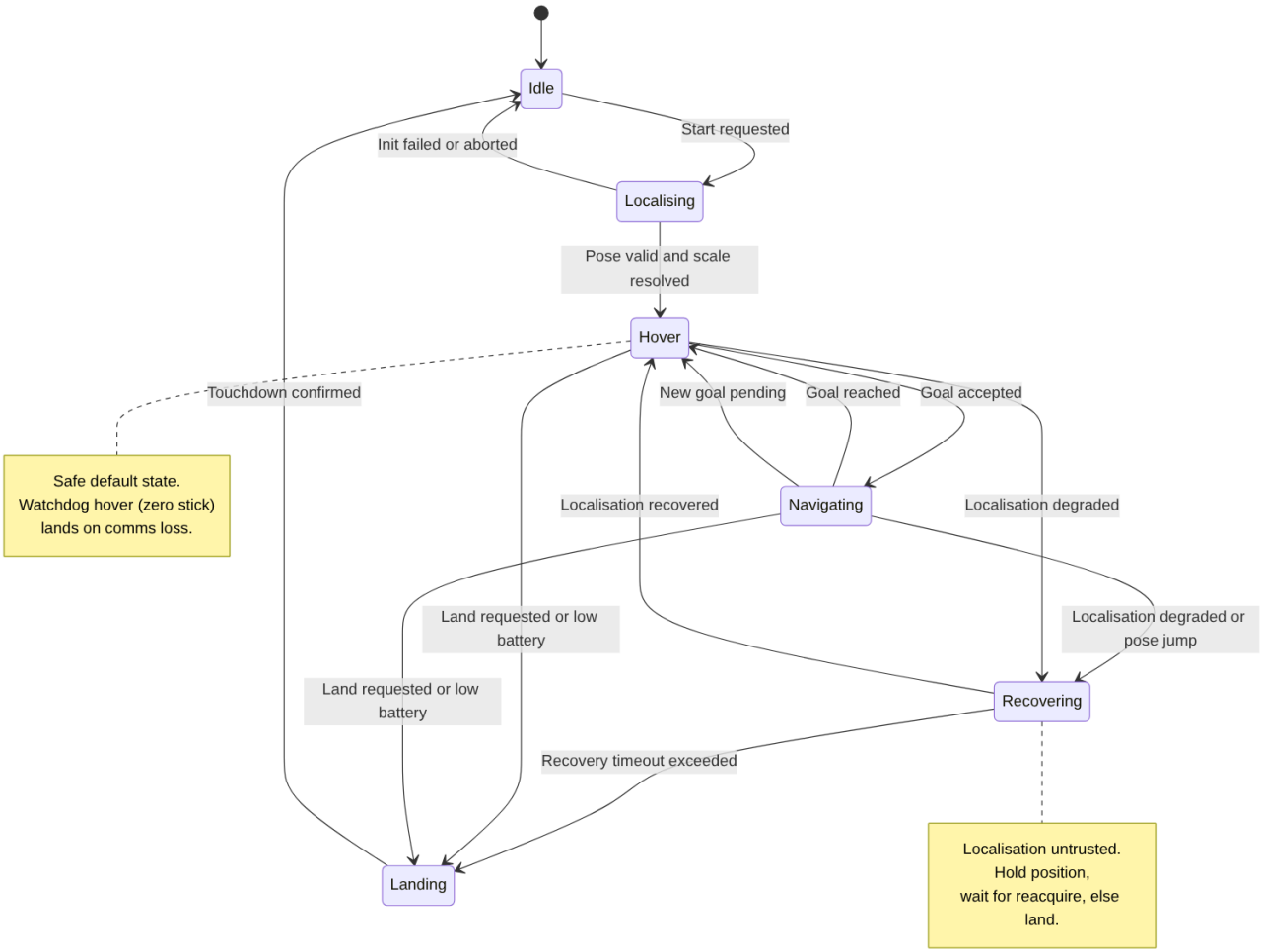
7a. Command authority (manual vs autonomous)

The RC arbitrates in hardware — manual override always wins. This machine records that authority, not the drone firmware's internal logic.



7b. Autonomous flight behaviour

Hover is the safe default that every failure drains into.



Thresholds and conditions to define

- Stack healthy: all flight-critical nodes up and pose valid.
- localisation "degraded": VO LOST, or VO_DISCONTINUITY, or telemetry transport gap (>500 ms with no packets), or sigma_s > SIGMA_S_MAX. Do NOT gate on pose covariance: p_x, p_y are observed by no update, so P_xx and P_yy grow monotonically and a COV_MAX threshold fires on any sufficiently long healthy flight.
- recovery timeout: 10 s before forcing Landing.
- low battery: 15 %.
- Stack healthy: warm clock offset tracker and aircraft airborne.

Cross-references

- Watchdog hover (300 ms comms loss) enters Hover in 7b and, on the phone side, is what the dead-man switch enforces regardless of stack state.

Project Conventions

1. Namespacing

All WildBridge telemetry/command topics are published under the `drone_1` namespace. The whole stack adopts this.

`/tf` and `/tf_static` are kept abosolute as per ROS standards.

---

2. Package structure

Packages map 1:1 to services from deployment.mmd

```
ros2_ws/
  src/
    drone_bringup/          # top-level launch + shared config
      launch/
        bridge_camera.launch.py
        system.launch.py    # includes all service launches
      config/
        common.yaml         # shared params
      package.xml
      CMakeLists.txt

    dji_controller/         # adopted from WildBridge
      dji_controller/
        submodules/
          dji_interface.py
        controller.py
      package.xml
      setup.py

    camera_streamer/        # python pkg
      camera_streamer/
        camera_decoder_node.py
      launch/
        camera_streamer.launch.py
      config/
        camera_streamer.yaml
      package.xml
      setup.py

    drone_localisation/     # python pkg
      drone_localisation/
        slam_node.py
        ekf_node.py
      launch/
        localisation.launch.py
      config/
        slam/params.yaml
        ekf/params.yaml
      package.xml
      setup.py

    vslam/
      config/
        camera_calibration.yaml
        slam_node.yaml
      src/
        slam_node.cpp
      launch/
        vslam.launch.py
```

```
    CMakeLists.txt
    package.xml

  drone_navigation/       # python pkg
    drone_navigation/
      trajectory_gen_node.py
      waypoint_follower_node.py
      position_controller_node.py
    launch/
      navigation.launch.py
    config/
      trajectory_gen/params.yaml
      waypoint_follower/params.yaml
      position_controller/params.yaml
    package.xml
    setup.py

  drone_evaluation/         # dev-only
    drone_evaluation/
      comparison_node.py
    launch/
      evaluation.launch.py
    config/
      comparison/params.yaml
      domain_bridge.yaml    # ros_domains_bridge config (0 -> 12)
    package.xml
    setup.py

  drone_interfaces/        # python pkg
    msg/
      AttitudeStamped.msg
      RelativeAltitudeStamped.msg
    CMakeLists.txt
    package.xml
```

3. Parameters

Two mechanisms, used together:

- **Declare** every parameter in the node with a default value.
- **Override** tuned values via `config/<node>/params.yaml`, loaded by launch.

---

4. Layered launch structure

- **Per-package launch** (`<service>.launch.py`): brings up that service's nodes, applies `namespace='drone_1'`, loads that package's param files. Runnable in isolation for development.
- **Top-level launch** (`drone_bringup/launch/system.launch.py`): includes the per-package launches. Brings up the whole system.

---

5. Run / build commands

Build (from `ros2_ws/`):

```
colcon build --symlink-install
source install/setup.bash
```

Run one service (dev):

```
ros2 launch drone_localisation localisation.launch.py
```

Run whole system:

```
ros2 launch drone_bringup system.launch.py
```

---

6. Docker compose commands

- **Dev default:** `command: ["bash"]` + `stdin_open: true` + `tty: true`.
- **Eventual per-service default:** the service's launch file, e.g. `command: ["ros2", "launch", "drone_localisation", "localisation.launch.py"]`.